

DSS5202 Sustainable Systems Analysis Assignment 3 Solutions

Notations:

We denote the four criteria and five alternative as follow:

$C_1 =$ Cooling effectiveness (long term)	$A_1 =$ Urban tree-canopy expansion
$C_2 =$ Financial feasibility	$A_2 =$ Cool-roof retrofit
$C_3 =$ Equity of public benefit	$A_3 =$ District cooling
$C_4 =$ Time to completion	$A_4 =$ Heat-health warning & response
	$A_5 =$ Public transit-stop shading

Question 1.

$$\mathbf{A} = \begin{matrix} & \begin{matrix} C_1 & C_2 & C_3 & C_4 \end{matrix} \\ \begin{matrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{matrix} & \begin{bmatrix} 1 & 3 & 2 & 5 \\ 1/3 & 1 & 1/2 & 2 \\ 1/2 & 2 & 1 & 3 \\ 1/5 & 1/2 & 1/3 & 1 \end{bmatrix} \end{matrix} \quad \mathbf{RGM} = \begin{bmatrix} 2.34035 \\ 0.75984 \\ 1.31607 \\ 0.42729 \end{bmatrix} \quad \mathbf{w} \approx \begin{bmatrix} 0.4832 \\ 0.1569 \\ 0.2717 \\ 0.0882 \end{bmatrix}$$

Using all the rows to estimate λ_{max} :

$$\mathbf{Aw} = \begin{bmatrix} 1.9383 \\ 0.6302 \\ 1.0917 \\ 0.3539 \end{bmatrix} \quad \lambda_{max} \approx \text{Average} \begin{pmatrix} 4.011557 \\ 4.017398 \\ 4.017844 \\ 4.011275 \end{pmatrix} = 4.0145$$

$$CI = \frac{4.0145 - 4}{4 - 1} = 0.00484 \quad CR = \frac{0.00484}{0.9} = 0.00538 < 0.1$$

Criterion C_1 , Cooling effectiveness (long term), has the highest priority, with a weight of 0.4832.

Criterion C_4 Time to completion has the lowest priority, with a weight of 0.0882.

Question 2.

Preference orderings from the pairwise comparison matrix:

$$C_1 \succ C_2, \quad C_1 \succ C_3, \quad C_1 \succ C_4, \quad C_2 \succ C_4, \quad C_3 \succ C_2$$

There are no preference cycles. The matrix in Question 1 satisfies the preference transitivity property.

Question 3.

Computing criteria weights with respect to Goal using exact method:

	C_1	C_2	C_3	C_4	Weight
C_1	1	3	2	5	0.482886
C_2	1/3	1	1/2	2	0.156990
C_3	1/2	2	1	3	0.271974
C_4	1/5	1/2	1/3	1	0.088150

$$\lambda_{\max} = 4.014521 \quad CI = 0.004840 \quad CR = 0.005378 < 0.1$$

Computing alternative weights with respect to criterion C_1 :

	A_1	A_2	A_3	A_4	A_5	Weight
A_1	1	1	2	4	3	0.333614
A_2	1	1	1	3	2	0.251500
A_3	1/2	1	1	3	2	0.218485
A_4	1/4	1/3	1/3	1	1/2	0.074303
A_5	1/3	1/2	1/2	2	1	0.122098

$$\lambda_{\max} = 5.052510 \quad CI = 0.013127 \quad CR = 0.011721 < 0.1$$

Computing alternative weights with respect to criterion C_2 :

	A_1	A_2	A_3	A_4	A_5	Weight
A_1	1	1/2	2	1/3	1	0.135173
A_2	2	1	3	1/2	1	0.211156
A_3	1/2	1/3	1	1/6	1/2	0.071078
A_4	3	2	6	1	3	0.426469
A_5	1	1	2	1/3	1	0.156124

$$\lambda_{\max} = 5.039377 \quad CI = 0.009844 \quad CR = 0.008790 < 0.1$$

Computing alternative weights with respect to criterion C_3 :

	A_1	A_2	A_3	A_4	A_5	Weight
A_1	1	2	1/3	1	1	0.190177
A_2	1/2	1	2	1	1	0.195094
A_3	3	1/2	1	1/2	1/2	0.190676
A_4	1	1	2	1	1	0.212026
A_5	1	1	2	1	1	0.212026

$$\lambda_{\max} = 5.615697 \quad CI = 0.153924 \quad CR = \mathbf{0.137432} > 0.1$$

Computing alternative weights with respect to criterion C_4 :

	A_1	A_2	A_3	A_4	A_5	Weight
A_1	1	1/2	2	1/5	1/2	0.098332
A_2	2	1	4	1/2	1	0.205466
A_3	1/2	1/4	1	1/9	1/4	0.050144
A_4	5	2	9	1	2	0.440593
A_5	2	1	4	1/2	1	0.205466

$$\lambda_{\max} = 5.005543 \quad CI = 0.001386 \quad CR = 0.001237 < 0.1$$

Computing the alternative global weights with respect to the goal:

$$\text{Global } \mathbf{w} = \begin{bmatrix} 0.333614 & 0.135173 & 0.190177 & 0.098332 \\ 0.251500 & 0.211156 & 0.195094 & 0.205466 \\ 0.218485 & 0.071078 & 0.190676 & 0.050144 \\ 0.074303 & 0.426469 & 0.212026 & 0.440593 \\ 0.122098 & 0.156124 & 0.212026 & 0.205466 \end{bmatrix} \begin{bmatrix} 0.482886 \\ 0.156990 \\ 0.271974 \\ 0.088150 \end{bmatrix} = \begin{bmatrix} 0.242710 \\ 0.225768 \\ 0.172941 \\ 0.199335 \\ 0.159247 \end{bmatrix}$$

The alternatives sorted from highest to lowest priority:

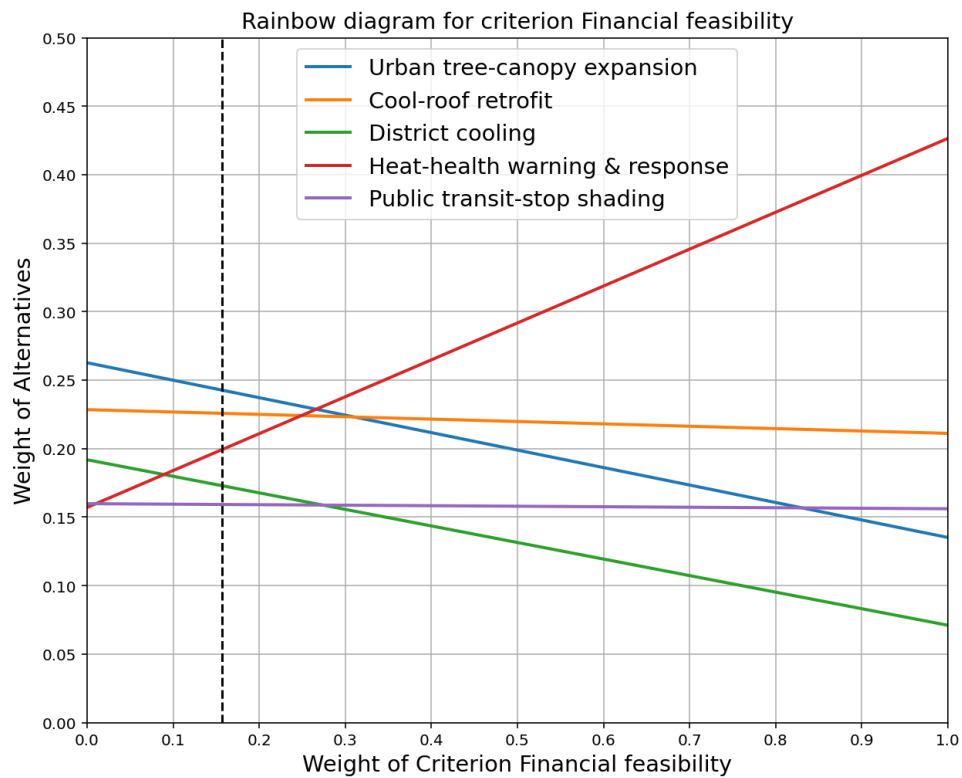
	Head-resilience measure	Weight
1	Urban tree-canopy expansion (A_1)	0.242710
2	Cool-roof retrofit (A_2)	0.225768
3	Heat-health warning & response (A_4)	0.199335
4	District cooling (A_3)	0.172941
5	Public transit-stop shading (A_5)	0.159247

Question 4.

The pairwise comparison matrix for the alternatives w.r.t. criterion C_3 (Equity of public benefit) has $CR = 0.137432 > 0.1$. Its inconsistency is unacceptable.

Question 5.

From the rainbow diagram when the priority weight of the criterion “Financial feasibility (C_2)” is varied from 0 to 1, the alternative “Heat-health warning & response (A_4)” has the highest priority when the value is between 0.27 to 1.



Question 6.

From the rainbow diagram when the priority weight of the criterion “Cooling effectiveness (long term) (C_1)” is varied from 0 to 1, the alternative “Cool-roof retrofit (A_2)” is never the highest priority. Hence the range of value is the empty set $\emptyset$.

